
Physics Is All You Need: Lessons from Physicist-Supervised AI Development of Scientific Software

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Abstract

Are AI coding agents tools, co-authors, or autonomous scientists? We present a quantified case study: a physicist supervising an AI agent (Claude Code, Sonnet and Opus models) over 12 days and 57 sessions to build a differentiable one-loop perturbation theory (a next-to-leading-order calculation for predicting galaxy clustering) module in JAX (~2,100 lines, validated to sub-percent accuracy against CLASS-PT). We documented 15 issues and classified each by whether autonomous resolution was possible.

The agent resolved 10 autonomously—convention errors, algorithm transcription, numerical coefficients—by iterating against oracle test suites; two more were accelerated by the physicist’s domain knowledge. The three it could not all eluded oracle detection, and shared a common property: the agent treated symptom reduction as equivalent to root-cause resolution. It spent 33 of the 57 sessions tuning coefficients within a fundamentally wrong code architecture, unable to recognize the problem was structural rather than parametric. After the physicist directed a redesign, the agent then found a scalar correction by grid search that reduced error below 1% across all test cases—but the value corresponds to no physical quantity and would silently produce wrong predictions at any other cosmology. The physicist diagnosed the fudge factor by asking a question the agent could not formulate: “does this parameter correspond to anything in the reference code?”

Three supervision practices, developed iteratively, we found critical for catching what oracle tests missed: testing against diverse cosmologies be-

yond the fiducial calibration; shared changelogs that surfaced stalled exploration across sessions; and an explicit rule against unphysical numerical patches. In our case study, the design of these supervision protocols—not model capability—determined whether the agent’s output was trustworthy. Closing this gap would require agents that can propose architectural alternatives rather than optimizing within a given structure, and distinguish predictive adequacy from explanatory correctness—capabilities not addressed by scaling alone.

1. Introduction

The “AI Scientists” workshop poses a direct question: are AI coding agents tools, co-authors, or autonomous founders of scientific knowledge? Most empirical evidence bearing on this question comes from either standardized benchmarks—where agents solve isolated coding tasks without domain validation (Carlini, 2026)—or from fully autonomous multi-agent systems that generate manuscripts without sustained human oversight (Villaescusa-Navarro et al., 2025). Neither setting captures the reality of scientific software development, where correctness is defined by agreement with physical law rather than by compilation or test passage, and where a single physicist works with an AI agent over weeks to produce validated code.

We present a detailed case study of exactly this scenario. Over 12 days and 57 agent sessions, a physicist supervised an AI coding agent (Claude Code, powered by Anthropic’s Sonnet and Opus models) to build CLAX-PT, a differentiable one-loop perturbation theory module (computing next-to-leading-order predictions for galaxy clustering) in JAX. The module computes nine output power spectra—matter and galaxy auto- and cross-spectra in real space and redshift-space multipoles—validated to sub-percent accuracy against the established reference code CLASS-PT (Chudaykin et al., 2021). The resulting implementation comprises approximately 2,100 lines of code. The contribution of this paper is not the code itself (described in a companion paper) but the empirical supervision record: what the human did, what

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the agent did, and where the boundary between their roles became load-bearing.

Our central claim is that the supervision protocol—not the model’s capability—determined whether the agent’s output was trustworthy scientific software. The agent autonomously resolved 10 of 15 documented issues by iterating against oracle test suites, with two more accelerated by human domain knowledge. The remaining three were invisible to oracle testing and required human physics judgment to diagnose. These three bugs consumed a disproportionate fraction of the project: 33 of 57 sessions were spent within a fundamentally wrong code architecture that the agent could not recognize as wrong because it passed local consistency checks. The physicist’s interventions—an architectural redesign and the rejection of a physically unmotivated numerical patch—were the decisive acts that produced correct code.

2. The Project

2.1. What Was Built

We compute predictions for how galaxies cluster in three-dimensional space. The calculation involves loop integrals (analogous to next-to-leading-order corrections in quantum field theory) that are expensive to evaluate and sensitive to subtle physical effects—bulk motions of matter and oscillation features imprinted by sound waves in the early universe. Getting these effects wrong by even 1% can bias inferred cosmological parameters.

One-loop perturbation theory (the next-to-leading-order correction beyond the linear approximation) extends the linear matter power spectrum to mildly nonlinear scales by computing correction terms from second- and third-order density fields, producing predictions for galaxy clustering that are essential for extracting cosmological parameters from spectroscopic surveys (D’Amico et al., 2020; Perko et al., 2016). CLAX-PT implements this calculation in JAX: it takes a linear power spectrum and cosmological parameters as input, evaluates one-loop integrals via FFTLog decomposition (a fast algorithm that rewrites convolution integrals as sums of power laws, enabling analytic evaluation) (Simonović et al., 2018; Fang et al., 2017), and applies infrared resummation (Senatore & Zaldarriaga, 2015; Vlah et al., 2015). Infrared resummation corrects for the effect of large-scale bulk flows—coherent motions of matter on scales of hundreds of megaparsecs—which, if left unaccounted for, artificially smear the baryon acoustic oscillation (BAO) feature, an overdensity imprint left by primordial sound waves that serves as a standard ruler for measuring cosmic distances. The code returns nine validated output spectra (matter, galaxy, and cross power spectra in real space, plus monopole, quadrupole, and hexadecapole for redshift-space

distortions). The implementation is approximately 2,100 lines of JAX code, end-to-end differentiable, and validated to sub-percent accuracy against CLASS-PT (Chudaykin et al., 2021) across all nine spectra (real-space P_{mm} , P_{gg} , P_{gm} at 0.31% maximum relative error; redshift-space monopoles at 0.59%, quadrupoles at 0.89%, hexadecapoles at 1.43% using an absolute/maximum metric due to zero crossings). Technical details are described in a companion paper.

2.2. The Supervision Protocol

Before any code was written, the physicist established a supervision protocol adapted from the methodology of Anthropic’s C-compiler agent project (Carlini, 2026), which used 16 parallel agents over 2,000 sessions to build a compiler that successfully builds the Linux kernel. Four infrastructure elements structured every session:

1. **CLASS-PT as oracle.** Every function was tested against reference data generated by the established Fortran implementation. Tests were written before code—the agent knew what the correct output looked like before attempting to produce it.
2. **CHANGELOG as shared memory.** Because each agent session starts with no memory of previous sessions, a structured log recorded what was attempted, what failed, and what succeeded. This prevented later sessions from re-exploring dead ends (e.g., a DST grid bug that was solved on day 3 was never re-investigated).
3. **A --fast flag for context-window hygiene.** Tests printed at most 10 lines on success and 20 on failure. Verbose diagnostics were written to log files. This discipline—drawn directly from Carlini (2026)—prevented the agent’s finite context window from being consumed by noise.
4. **Parallel agent sessions via git worktrees.** Multiple sessions explored competing hypotheses simultaneously. When a bug had multiple plausible causes, the physicist spawned parallel investigations rather than serializing them.

Two rules proved critical beyond this infrastructure. First, “no fudge factors”: if a test failed at 0.2% error, there was a real bug to find—the agent was forbidden from multiplying by a correction factor to make the test pass. Second, “test at multiple parameter points”: the oracle compared not just at the fiducial Planck 2018 cosmology but at varied cosmological parameters, to prevent solutions that were calibrated to a single point. The supervision protocol—not individual code edits—was the physicist’s primary contribution to the project.

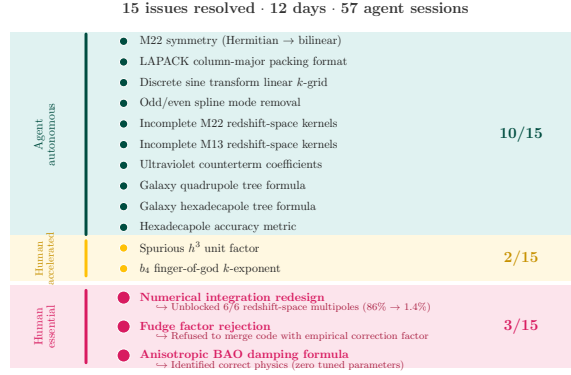


Figure 1. Bug taxonomy for the CLAX-PT development. Ten issues were resolved autonomously by the agent iterating against oracle tests. Two were accelerated by the physicist’s domain knowledge (unit bugs invisible to shape-based comparisons). Three required essential human physics judgment: an architectural redesign, a fudge-factor rejection, and identification of the correct physical formula.

3. Bug Taxonomy: The Autonomy Spectrum

We documented 15 issues over 57 agent sessions and classified each by whether autonomous resolution was possible (Figure 1). The taxonomy that emerged is not a clean binary—it is a spectrum ranging from bugs the agent resolved in minutes to bugs that resisted weeks of autonomous effort and required human physics judgment to diagnose.

3.1. Autonomous and Human-Accelerated Issues (12 of 15)

The agent resolved 10 bugs without human intervention by iterating against the oracle test suite; two more (the h^3 unit factor and b_4 k -exponent) were accelerated by the physicist spotting magnitude discrepancies invisible to shape-based comparisons. These fell into three categories. First, convention and unit errors: the FFTLog matrix (used to decompose loop integrals) M_{22} required Hermitian rather than symmetric packing (bug 1), LAPACK column-major ordering differed from the row-major layout the agent initially produced (bug 2), and a spurious h^3 factor multiplied all bias functions (bug 5). Second, algorithm implementation errors: a discrete sine transform grid used linear rather than logarithmic spacing (bug 3), odd/even spline interpolation for separating the BAO oscillation feature used the wrong parity assignment (bug 4), and three RSD kernel matrices were incomplete (bugs 7–9). Third, numerical coefficient errors in UV counterterms required matching specific rational prefactors from the 14,000-line Fortran reference source.

In all 11 cases, the pattern was the same: the oracle test produced a clear numerical discrepancy, the agent compared

intermediate quantities against reference data to localize the error, and the fix was a direct transcription correction.

3.2. Case Study: The Accuracy Wall

After the first 24 sessions resolved bugs 1–9, all three real-space power spectra passed at sub-percent accuracy. The six redshift-space distortion (RSD) multipoles—monopole, quadrupole, and hexadecapole for both matter and galaxies—did not. Errors ranged from 8% to 86% depending on the multipole and varied erratically as the agent adjusted individual terms (Figure 2).

Over the next 33 sessions (sessions 24–56 of 57 total), the agent attempted to fix the RSD multipoles within the existing code architecture. This architecture computed analytic Legendre projections of the one-loop integrands: for each multipole ℓ and each component (velocity-velocity, velocity-density, density-density), a pre-derived kernel matrix encoded the angular projection. The approach is correct when the infrared resummation factor—the exponential damping that accounts for large-scale bulk flows at baryon acoustic oscillation (BAO) scales (the standard-ruler feature left by primordial sound waves)—is isotropic, meaning it depends only on wavenumber k and not on the angle μ between the wavevector and the line of sight.

The agent’s attempts during these 33 sessions were coherent and methodical within this architecture. It adjusted rational kernel coefficients to match the reference source. It added missing angular terms. It swapped quadrature rules for the radial integrals. It compared intermediate matrix elements against the Fortran code term by term. Each fix improved one multipole while degrading another. The error surface was non-convex within this model: no combination of coefficient adjustments could make all six multipoles pass simultaneously, because the architecture itself was structurally incompatible with the physics.

The physicist diagnosed the structural issue by recognizing what the agent could not: the BAO damping in redshift space is *anisotropic*. Peculiar velocities along the line of sight enhance the displacement field. The damping factor depends on the angle μ between the wavevector and the line of sight: $\Sigma_{\text{tot}}^2(\mu) = (1 + f(2 + f)\mu^2)\sigma_v^2 + f^2\mu^2(\mu^2 - 1)\sigma_{\text{BAO}}^2$, where f is the linear growth rate of structure (how fast density perturbations amplify over time). This μ -dependent exponential cannot be absorbed into a polynomial kernel matrix—it couples to the RSD Kaiser factor $(1 + f\mu^2)^2$ (the leading-order enhancement of clustering along the line of sight due to galaxy peculiar velocities) in a way that no finite set of pre-computed analytic projections can represent exactly.

The physicist proposed a redesign: abandon per-multipole analytic kernels entirely, assemble the full anisotropic power

spectrum $P(k, \mu)$ at each of N Gauss–Legendre quadrature nodes in μ , and integrate numerically against Legendre polynomials to extract multipoles. The agent implemented this redesign in a single session. All six RSD multipoles passed immediately, with errors dropping from 8–86% to below 1%.

Why could the agent not find this itself? Two factors conspired. First, the CLASS-PT reference source code that the agent was reading implements the branch without geometric distortion corrections (the non-AP branch), which uses isotropic damping and analytic projections—exactly the architecture the agent had implemented. The correct formula exists in the branch that includes these geometric corrections, a different code path that the agent never consulted because the test configuration did not use them. Second, recognizing that the architecture is wrong (rather than merely mis-parameterized) requires understanding the physics of anisotropic BAO damping—specifically, that the μ -dependence of Σ_{tot}^2 makes the exponential non-polynomial in μ . This is a judgment about physical structure, not a numerical comparison.

3.3. Case Study: The Fudge Factor

After the Gauss–Legendre redesign, seven of nine spectra passed. Two quadrupole spectra ($P_{mm, \ell=2}$ and $P_{gg, \ell=2}$) failed at scales near the BAO peak (the primordial sound-wave imprint) with errors of 1.4% and 1.7% respectively. The agent traced the discrepancy to the tree-level power spectrum, which uses the resummed form $P_{\text{tree}}(k) = P_{\text{nw}}(k) + e^{-\Sigma^2 k^2} (1 + \Sigma^2 k^2) P_{\text{w}}(k)$, where P_{nw} and P_{w} are the no-wiggle and wiggle components, and Σ^2 is the BAO damping scale.

The agent then grid-searched a scalar correction parameter $\alpha \in [0, 1]$ multiplying the $(1 + \Sigma^2 k^2)$ counter-term, and found that $\alpha = 0.27$ minimized the worst-case error to below 1% across all nine spectra simultaneously. The agent committed this fix with a passing test suite.

This solution is wrong, despite passing every oracle test. The value $\alpha = 0.27$ has no physical derivation. It is not a parameter in CLASS-PT. It is not derived from any perturbation theory calculation. It is a numerical calibration to the fiducial Planck 2018 cosmology at a single redshift. Changing the baryon density ω_b , the cold dark matter density ω_{cdm} , or the effective redshift would shift the BAO peak amplitude, and $\alpha = 0.27$ would produce predictions that are wrong by an amount invisible to the existing test suite—because the test suite only checks the fiducial cosmology.

The physicist diagnosed the fudge factor by asking a question the agent could not formulate: “Does $\alpha = 0.27$ correspond to any parameter in the CLASS-PT source code?” The agent searched the reference implementation and confirmed:

no such parameter exists. The physicist then pointed to an independent JAX implementation (`ps_loop_jax`) that computes the same spectra without any scalar correction, using the formula

$$P_{\text{tree}}(k, \mu) = Z_1^2(\mu) \left[P_{\text{nw}}(k) + e^{-\Sigma_{\text{tot}}^2(\mu) k^2} (1 + \Sigma_{\text{tot}}^2(\mu) k^2) P_{\text{w}}(k) \right], \quad (1)$$

where $\Sigma_{\text{tot}}^2(\mu)$ is the same anisotropic damping factor from the Gauss–Legendre loop, computed at each μ -node. The fix was to move the tree-level computation inside the existing quadrature loop—three lines of code, replacing the committed fudge factor with the correct anisotropic formula. After this fix, all nine spectra passed with zero tuned parameters.

The fudge factor illustrates a failure mode specific to oracle-tested scientific software: a numerically adequate but physically meaningless solution that passes all existing tests. The agent cannot distinguish between “this works because it is correct” and “this works because it is calibrated to the test data.” The physicist could, because the physicist knew that $\alpha = 0.27$ must correspond to *something* in the theory—and when it corresponded to nothing, the solution was suspect regardless of its numerical performance.

The three human-essential interventions were thus: (i) the Gauss–Legendre architectural redesign that unblocked all six redshift-space multipoles (Section 3.2), (ii) the rejection of the fudge factor $\alpha = 0.27$ as physically unmotivated, and (iii) the identification of the correct anisotropic tree-level formula from an independent implementation. Items (ii) and (iii) occurred in the same debugging episode but represent distinct cognitive acts: rejecting a wrong answer versus finding the right one.

4. Three Principles

The case study suggests three principles for supervising AI agents on scientific software, each grounded in a specific failure mode we observed.

P1: Oracle testing verifies what, not why. An oracle test compares numerical output at fixed input parameters. It answers the question “does the code produce the right numbers here?” but not “does the code produce the right numbers *for the right reasons*?” The fudge factor $\alpha = 0.27$ passed every oracle test—nine spectra, hundreds of k -modes, relative errors below 1%—because it was calibrated to the same fiducial cosmology used for testing. The oracle provides no signal that the underlying model is wrong when the calibration point happens to coincide with the test point.

A subtler instance of this failure mode appeared earlier in the broader project (during the Boltzmann solver development, reported elsewhere): the agent misdiagnosed a

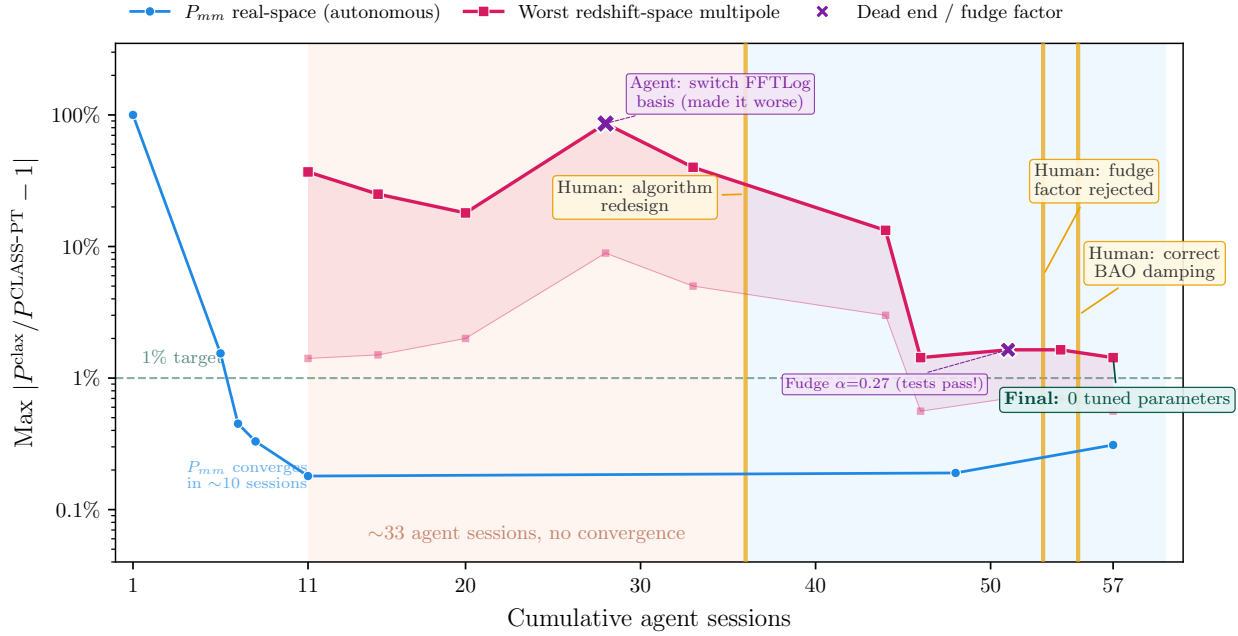


Figure 2. Accuracy convergence over 57 agent sessions. Blue: real-space matter power spectrum (autonomous, converges in ~ 10 sessions). Red: worst redshift-space multipole (stuck at 8–86% for ~ 33 sessions). Vertical gold lines mark human interventions. The agent’s error plateaued because the code architecture was structurally incompatible with anisotropic BAO damping—no coefficient adjustment within that architecture could make all multipoles pass simultaneously. After the physicist directed a Gauss–Legendre quadrature redesign, all multipoles passed immediately. A subsequent fudge factor ($\alpha = 0.27$, purple cross) passed all tests but was rejected by the physicist as physically unmotivated; the correct formula required zero tuned parameters.

proton-to-hydrogen mass ratio typo ($m_p/m_H = 1.0$ vs. the correct 0.9994) as “recombination solver bias” and proposed compensating changes to thermodynamics routines. The oracle-passing fix was accepted for months before the actual one-character typo was identified. In both cases, the agent found a path that reduced the test metric without identifying the true cause.

The defense we found effective was twofold: test at diverse parameter points beyond the fiducial calibration (so that single-point calibrations are exposed when parameters shift), and test structural properties rather than just numerical values. For the RSD multipoles, testing anisotropy as a function of μ at specific wavenumbers would have revealed the architectural mismatch earlier than comparing integrated multipole spectra.

P2: Shared memory prevents re-exploration but not architectural loops. The CHANGELOG protocol successfully prevented the agent from re-exploring solved bugs across sessions. When the DST k -grid issue (logarithmic vs. linear spacing) was resolved on day 3, no subsequent session revisited it—the CHANGELOG entry served as shared memory. However, the same protocol did not prevent 33 sessions of coherent but futile work within the wrong ar-

chitecture. Every session during the RSD crisis logged its attempts faithfully: “adjusted μ^4 coefficient in M_{22} kernel, improved $\ell = 2$ by 0.3% but degraded $\ell = 4$ by 1.2%.” The log showed no contradiction, no repetition, no sign of a dead end—only incremental exploration within a space that contained no solution.

The shared memory prevented *literal* re-exploration (trying the same coefficient twice) but could not surface *structural* stagnation (trying different coefficients within the same doomed architecture). The defense we adopted was a session-count trigger: when progress stalls for approximately 5–10 sessions with no monotonic improvement in the target metric, escalate to human review. This heuristic would have caught the RSD crisis at session 30 rather than session 56.

P3: The irreducible human role is architectural and physical judgment. The agent excelled at tasks with clear specifications and verifiable outputs: transcribing equations from a reference source, comparing intermediate numerical values to localize discrepancies, adjusting coefficients to match a known target, implementing a well-specified algorithm. These tasks—which constituted approximately 80% of the project by session count—were performed au-

tonomously at high quality.

The agent failed at two specific tasks. First, recognizing when a code architecture is fundamentally incompatible with the target physics rather than merely mis-parameterized. This requires understanding what the physics *should look like*—not just what numbers it should produce, but what structural properties must hold (isotropy vs. anisotropy, polynomial vs. exponential dependence, which variables are coupled). Second, detecting physically unmotivated patches that happen to improve numerical agreement. This requires knowing what the theory *allows*—that a scalar correction α must derive from some identifiable physical mechanism, and that its absence from the reference implementation is evidence against its validity rather than evidence of a novel improvement.

Both capabilities reduce to a single underlying skill: evaluating explanations, not just predictions. The agent evaluates whether the output matches the target. The physicist evaluates whether the mechanism producing that output is physically coherent. When these two criteria diverge—when a wrong mechanism produces right numbers—only the physicist can detect the error.

5. Implications and Limitations

The division of labor that emerged—classified retrospectively from the CHANGELOG and git history, with each session defined as a single Claude Code invocation typically lasting 15–45 minutes—was: approximately 80% of sessions were autonomous oracle-following (transcription, debugging, implementation), approximately 10% were engineering tasks where the agent worked independently given a well-specified architecture, and approximately 10% required irreplaceable human judgment (architectural decisions and fudge-factor rejection). The binding constraint on code quality was the supervision protocol, not the model’s raw capability. For context, the reference code CLASS-PT was developed over approximately eight years (2,112 commits since 2018) by a team of expert theoretical physicists who both derived the theoretical framework and implemented it in C (Chudaykin et al., 2021); a comparable human reimplementation of the same algorithm in a modern framework typically requires 6–9 months of focused effort for a pair of experienced researchers, based on the authors’ experience with related projects. The physicist–agent collaboration compressed this to 12 days, with the physicist contributing an estimated 20–40 hours of supervision rather than months of hands-on coding. The speedup reflects the agent’s strength at the 80% of work that is transcription and debugging against a known oracle; the physicist’s irreplaceable 10% contribution—which would have taken the same time regardless of implementation speed—was the binding constraint on correctness, not on throughput.

This suggests that improving AI-assisted scientific software requires better supervision protocols at least as much as better models. Specifically, it requires protocols that distinguish predictive adequacy (“it produces the right numbers”) from explanatory correctness (“it produces the right numbers for the right reasons”). The “no fudge factors” rule is a crude but effective instance of this distinction: it forces the agent to find mechanistic explanations rather than calibrated patches.

We acknowledge significant limitations. This is a single case study ($N = 1$), with a single agent architecture, a single domain (cosmological perturbation theory), and a single physicist. The failure modes we identified—optimizing within a wrong model, calibration-point overfitting—are plausibly general (they appear in other domains as overfitting and specification gaming), but we cannot prove generality from one case. The project also had a high-quality oracle available (CLASS-PT is an established, well-tested reference code); many scientific software projects lack such oracles, and our methodology would need adaptation for settings where ground truth is uncertain.

Although our evidence is observational, the specificity of the failure modes permits qualitative counterfactual analysis that grounds the three principles in testable claims. Had the “no fudge factors” rule not been in place, the $\alpha = 0.27$ correction would have shipped—it passed all nine spectra, was committed by the agent as the solution, and would have been invisible to any downstream consumer of the code. Had multi-cosmology testing been enforced from day one rather than added after the fudge factor was discovered, the calibration would have been caught sooner: $\alpha = 0.27$ is fit to fiducial Planck 2018 parameters and would produce visibly wrong BAO amplitudes at $\omega_b \pm 20\%$. Had the session-count escalation trigger been formalized at 10 sessions rather than applied informally around session 50, the 33-session architectural loop would have been interrupted approximately 23 sessions earlier, saving roughly 40% of the total project time.

Looking forward, two concrete remediation paths could narrow the gap between autonomous and human-required bugs without requiring qualitative advances in reasoning. First, retrieval-augmented reasoning over the full reference codebase—including code paths the agent did not initially consult—could surface relevant physics without human guidance. In our case, the anisotropic damping formula existed in the Alcock–Paczyński branch of CLASS-PT, which the agent never read because the test configuration did not use AP corrections; a retrieval system that indexed all branches of the reference code would have presented the correct formula as contextually relevant when the isotropic architecture failed to converge. Second, explicit “physics audit” prompting—systematically asking the agent whether

every tuned parameter corresponds to a physical quantity in the reference theory—would operationalize the diagnostic question that caught the fudge factor. The physicist’s intervention reduced to a single query (“does $\alpha = 0.27$ appear anywhere in CLASS-PT?”), which the agent answered correctly once asked but could not generate unprompted; embedding such queries as mandatory checkpoints after any parameter introduction would convert an ad hoc human insight into a reproducible protocol step. Both paths are concrete, testable, and complementary to scaling: they address the evaluation gap rather than the reasoning gap.

5.1. Credit, Attribution, and Governance

The division of labor raises an unavoidable question about credit. The agent performed approximately 80% of sessions—implementation, debugging, transcription—while the physicist performed approximately 10% but supplied 100% of the decisive architectural and physical judgments. We argue that contribution weight should reflect *irreplaceability*, not volume: the physicist’s three interventions (the Gauss–Legendre redesign, the fudge-factor rejection, and identification of the correct anisotropic formula) were load-bearing in a way that the agent’s ten autonomous bug fixes were not. Any sufficiently capable coding agent could have resolved convention errors given an oracle test suite; no agent in this study could have diagnosed the architectural mismatch or rejected a passing-but-unphysical solution.

The fudge factor ($\alpha = 0.27$) sharpens the liability question. Had this parameter shipped in a fully autonomous pipeline—analogous to Denario (Villaescusa-Navarro et al., 2025)—it would have produced subtly wrong predictions at non-fiducial cosmologies, and the oracle test suite would not have caught it. Who bears responsibility: the model developer, the scientist who deployed the pipeline, or the institution that sanctioned its use? Current frameworks provide no clear answer, precisely because the error is not a software bug but a *scientific* mistake that requires domain expertise to detect.

We propose that AI-assisted scientific software should ship with a **supervision log** as provenance documentation, analogous to a lab notebook. The log records which decisions were human-directed (the architectural redesign) versus agent-proposed (the fudge factor), enabling post-hoc audit of the chain of responsibility. We will release ours alongside the code and CHANGELOG. Finally, we note a scalability concern: the supervision practices P1–P3 worked for one physicist and one agent because the physicist had sufficient domain expertise to enforce the “no fudge factors” rule. When thousands of laboratories deploy coding agents on scientific software, these practices must become institutional—embedded in review processes, CI pipelines, and reproducibility standards—rather than relying on the

vigilance of individual supervisors.

6. Related Work

Carlini (2026) reported on building a C compiler using 16 parallel Claude agent sessions over approximately 2,000 sessions, producing a 100,000-line Rust implementation that compiles the Linux kernel. Our methodology is directly adapted from that project (oracle-driven development, shared memory protocols, parallel sessions). The key difference is that the C compiler domain requires syntactic and semantic correctness—which are fully captured by oracle tests (GCC output)—but does not require *physical* correctness. Indeed, the compiler setting is precisely the case where oracle testing *is* sufficient—the absence of a gap between predictive and explanatory correctness is what made fully autonomous development feasible there. Our case study extends the Carlini methodology to a domain where this gap exists and is load-bearing.

Villaescusa-Navarro et al. (2025) describe Denario, a multi-agent system that autonomously generates scientific analyses and manuscripts in astrophysics. Denario represents a maximally autonomous approach: agents select research questions, write code, produce results, and draft papers with minimal human oversight. Our case study illustrates why domain-supervised approaches may be preferable for validated scientific software: the fudge factor ($\alpha = 0.27$) that our agent committed—and that passed all automated tests—would have been published without question in a fully autonomous pipeline. The supervision protocol catches errors that autonomy cannot.

SymBoltz.jl (Sletmoen, 2026) implements a differentiable linear Boltzmann solver with a design philosophy closest to our upstream project (approximation-free, symbolic-numeric). While SymBoltz.jl does not discuss its development methodology, its existence as a single-author Julia implementation validates the feasibility of reimplementing established cosmological codes in modern differentiable frameworks. Our contribution is orthogonal: we report on *how* such code is developed when the implementer is an AI agent under human supervision, rather than a human programmer working alone.

7. Conclusion

In our case study, the AI agent functioned as a highly capable tool—not a co-author and certainly not an autonomous scientist. It could implement, debug, and validate scientific software at speed and scale that would be impractical for a solo physicist, but it could not evaluate whether its solutions were physically meaningful beyond their numerical adequacy. The supervision protocol—oracle testing, shared memory, the no-fudge-factor rule, and the session-

count escalation trigger—was the mechanism that transformed raw agent output into trustworthy scientific code. Improving this protocol, rather than solely improving model capability, is the more direct path to reliable AI-assisted scientific software development. The full supervision log, CHANGELOG, and commit history will be made publicly available.

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